SMART STUDY LIBRARY MANAGEMENT SYSTEM

ZenStudy

A centralized digital platform for managing memberships, study-space bookings, attendance, payments and library operations.

DeerInTheCode · Team-051

Software Engineering Project — May 2026

IITM Online BS Degree Program · Indian Institute of Technology, Madras

What We'll Cover

01 **Introduction & Team**
Who we are and what we built

02 **Vision & Users**
Who ZenStudy serves

03 **User Challenges & Needs**
The problem we identified

04 **Platform Architecture**
How the system works

05 **Smart Study Operations**
The major capabilities

06 **Implementation & Testing**
How we built and validated it

07 **Technology & Project Journey**
Our engineering process

08 **Conclusion & Impact**
What ZenStudy delivers

Welcome to ZenStudy

Study libraries are increasingly used by students who need dedicated, quiet spaces for focused learning.

However, many study libraries still depend on manual or semi-digital processes for:

- Membership management
- Attendance
- Walk-in admissions
- Complaints
- Seat reservations
- Payments
- No-show management
- Administrative reporting

ZenStudy brings these workflows together into one centralized digital platform.

One Platform for the Complete Study-Library Journey

Three workflows, connected through a common platform.

Students

Find → Book → Check In → Study → Check Out

Library Staff

Register → Verify → Allocate → Monitor → Resolve

Administrators

Configure → Manage → Monitor → Analyze

ZenStudy connects all three workflows through a common platform.

Meet the Team

DeerInTheCode · Team-051

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Diksha Sharma

Product Manager

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Karthik V N

Frontend Developer

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Neha Chacko

Scrum Master

GT

Gaurav Tomar

Backend Developer

AA

Akshatha A

Code Reviewer / Tester

Our Vision

Make study-space management simple, transparent and efficient.

1

Students

To independently manage their memberships, bookings and study sessions.

2

Library Staff

To efficiently manage daily operations.

3

Administrators

To control policies and monitor overall library performance.

Who Uses ZenStudy?

Three roles. One connected platform.

STUDENT

- Manage membership
- Check availability
- Book seats
- Modify/cancel bookings
- Check in/out
- View history
- Make complaints

LIBRARY STAFF

- Register students
- Verify memberships
- Manage bookings
- Allocate seats
- Handle check-ins
- Manage walk-ins
- Record payments
- Resolve complaints

ADMIN

- Manage staff
- Manage memberships
- Configure library
- Manage seats
- Publish announcements
- Monitor reports

Designed Around Real Workflows

1

Student Dashboard

Membership · Availability · Bookings · Attendance · Payments · Complaints

2

Staff Dashboard

Students · Bookings · Check-in · Walk-ins · Extensions · Payments · Complaints

3

Admin Dashboard

Staff · Memberships · Seats · Settings · Announcements · Reports

The Problem We Set Out to Solve

Students

- ✕ No convenient real-time seat visibility
- ✕ Difficult reservation management
- ✕ Limited membership visibility
- ✕ Manual extension requests
- ✕ Fragmented booking and payment history

Staff

- ✕ Manual registration
- ✕ Manual attendance
- ✕ Complex seat allocation
- ✕ Walk-in management
- ✕ Manual payment recording
- ✕ Complaint tracking

Administrators

- ✕ Limited occupancy insights
- ✕ Difficult membership monitoring
- ✕ Limited revenue visibility
- ✕ Manual policy management

From Problems to Solutions

User Challenge	ZenStudy Solution
Seat uncertainty	**Real-time availability**
Booking difficulty	**Online reservations**
Membership tracking	**Membership management**
Manual attendance	**Digital check-in/out**
Unused reserved seats	**No-show management**
Extension requests	**Extension workflow**
Manual payments	**Payment recording**
Complaints	**Centralized support**
Communication gaps	**Announcements**
Limited insights	**Reports & analytics**

ZenStudy — High-Level Architecture

Role-Based Frontend

Student Interface · Staff Interface · Admin Interface

↓

Application / API Layer

Authentication · Membership · Booking · Seat Allocation · Attendance · Payments · Complaints · Announcements · Reports

↓

Relational Database

Users · Students · Memberships · Seats · Bookings · Attendance · Payments · Complaints · Announcements · Settings

Seven Core Components

The building blocks of the ZenStudy platform.

01 Student

Booking · Membership · Self-Service

02 Library Staff

Operations · Attendance · Payments

03 Admin

Administration · Configuration · Analytics

04 Booking & Seat Allocation

Reservations · Availability · Reallocation

05 Attendance

Check-in · Check-out · Occupancy

06 Payments

Transactions · Receipts · History

07 Complaint & Communication

Support · Announcements · Notifications

Relational Data Architecture

- **Users**
  Role-based accounts
- **Students**
  Student records
- **Membership Plans**
  Available plans
- **Memberships**
  Student subscriptions
- **Seats**
  Available study spaces
- **Bookings**
  Reservations
- **Attendance**
  Check-in/check-out
- **Extension Requests**
  Additional study time
- **Payments**
  Financial records
- **Complaints**
  Issue management
- **Announcements**
  Communication
- **Library Settings**
  Operational configuration

Smart Seat & Booking Management

1. **Check Availability** — Students can identify available seats for a selected time.
2. **Reserve** — Book an available seat.
3. **Modify** — Change the booking within permitted rules.
4. **Cancel** — Cancel an existing reservation.
5. **Release** — Cancelled or no-show bookings release the seat.

Flexible Booking Extensions

1	**Student**	Requests additional study time.
↓		
2	**System**	Validates time, existing booking, extension limits, duplicate requests and ownership.
↓		
3	**Staff**	Reviews the request.
↓		
4	**Decision**	Approve or reject.
↓		
5	**Seat Conflict**	If required, an alternate available seat can be assigned.

Digital Check-In & Attendance

1	**Unique Check-In Code**	Each booking has a unique code.
2	**Staff Verification**	Staff validates the code at arrival.
3	**Attendance**	Check-in timestamp is recorded.
4	**Occupancy**	Seat status is updated.
5	**Checkout**	Session is completed and checkout time recorded.
6	**History**	Attendance remains available for future reference.

Recovering Unused Capacity

1	**Confirmed Booking**	A student holds a confirmed seat reservation.
2	**Student Does Not Arrive**	The student fails to check in.
3	**Grace Period**	A window is allowed before action is taken.
4	**No-Show Candidate**	The booking is flagged for review.
5	**Staff Marks No-Show**	Staff confirms the no-show status.
6	**Seat Released**	The seat is freed for other students.
7	**Seat Available for Rebooking**	The seat returns to the availability pool.

Complete Membership Management

Membership plan creation, modification, deletion and management were covered by Milestone 4 testing.

ADMIN

- Create plans
- Update plans
- Manage plan status
- Manage memberships
- Extend memberships

STAFF

- Register students
- Verify eligibility

STUDENT

- View membership
- View plan
- View validity

Centralized Payment Records

ZenStudy supports offline payment recording through Cash, UPI and Card.

1. **Staff records transaction**
2. **Payment stored**
3. **Student views payment history**
4. **Staff accesses student payment records**
5. **Admin uses information for reporting**

The implementation validates successful payments as well as invalid payment modes, missing amounts, negative amounts and role restrictions.

Cash **UPI** **Card**

Centralized Support

The implemented API set includes complaint creation, staff review and announcement management.

Complaints

1. Student submits complaint
2. Staff reviews complaint
3. Status: Open → In Progress → Resolved

Communication

1. Administration publishes announcements
2. Users view important updates

Operational Visibility

The system design identifies these as key administrative reporting areas.

1

Occupancy

Understand how study spaces are being utilized.

2

Memberships

Track membership statistics and status.

3

Attendance

Monitor study-library usage.

4

Revenue

Track payment-related information.

5

Configuration

Control operating hours, booking policies and capacity.

UI Screens

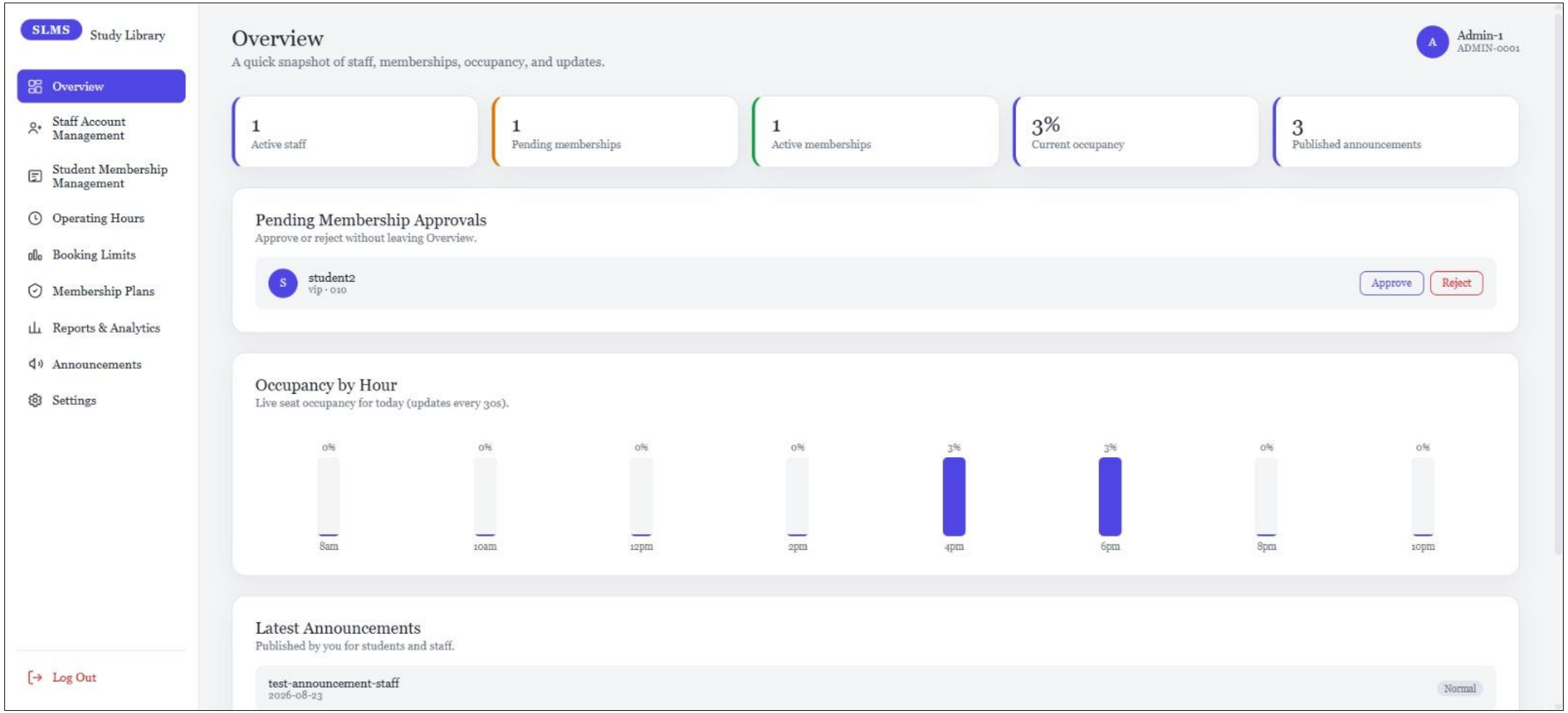


Fig: Admin Dashboard

UI Screens

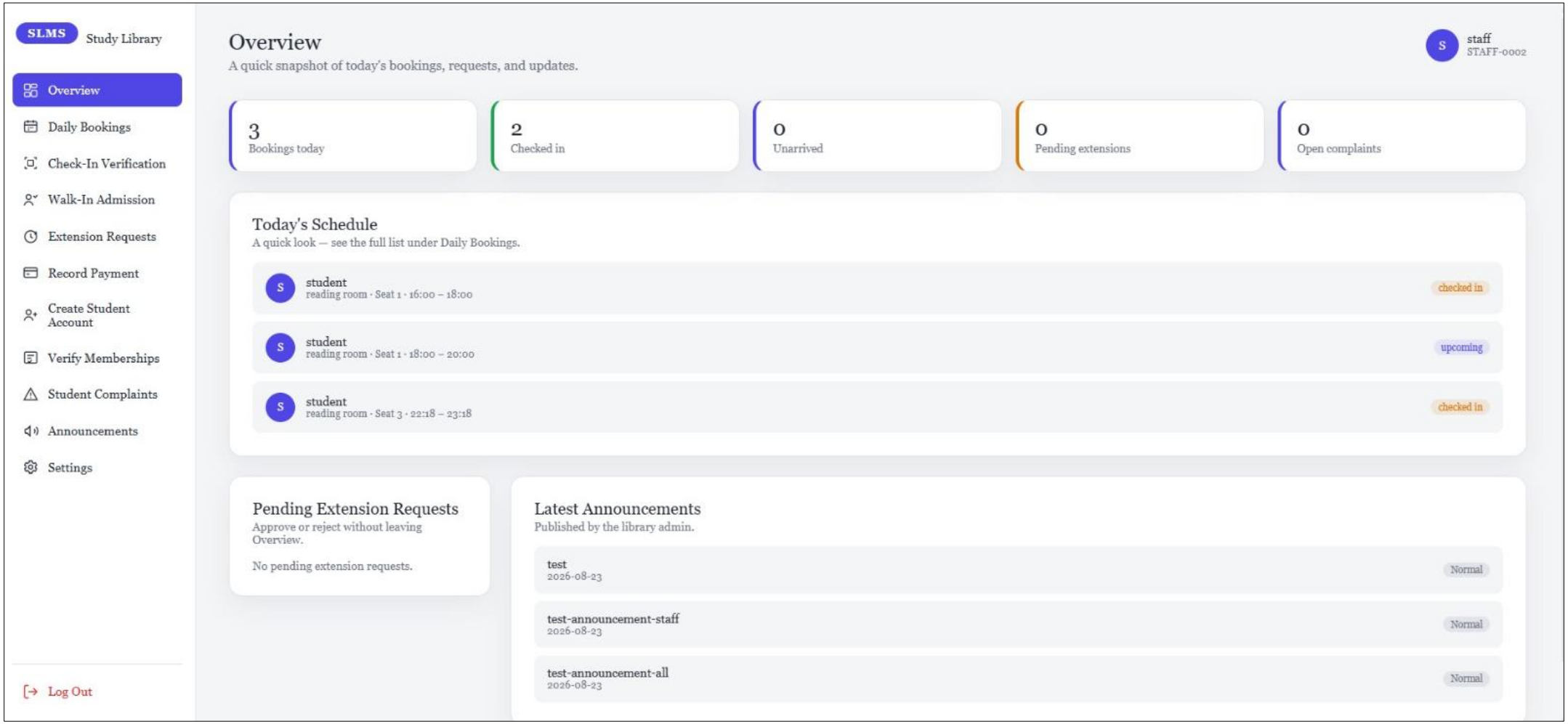


Fig: Staff Dashboard

UI Screens

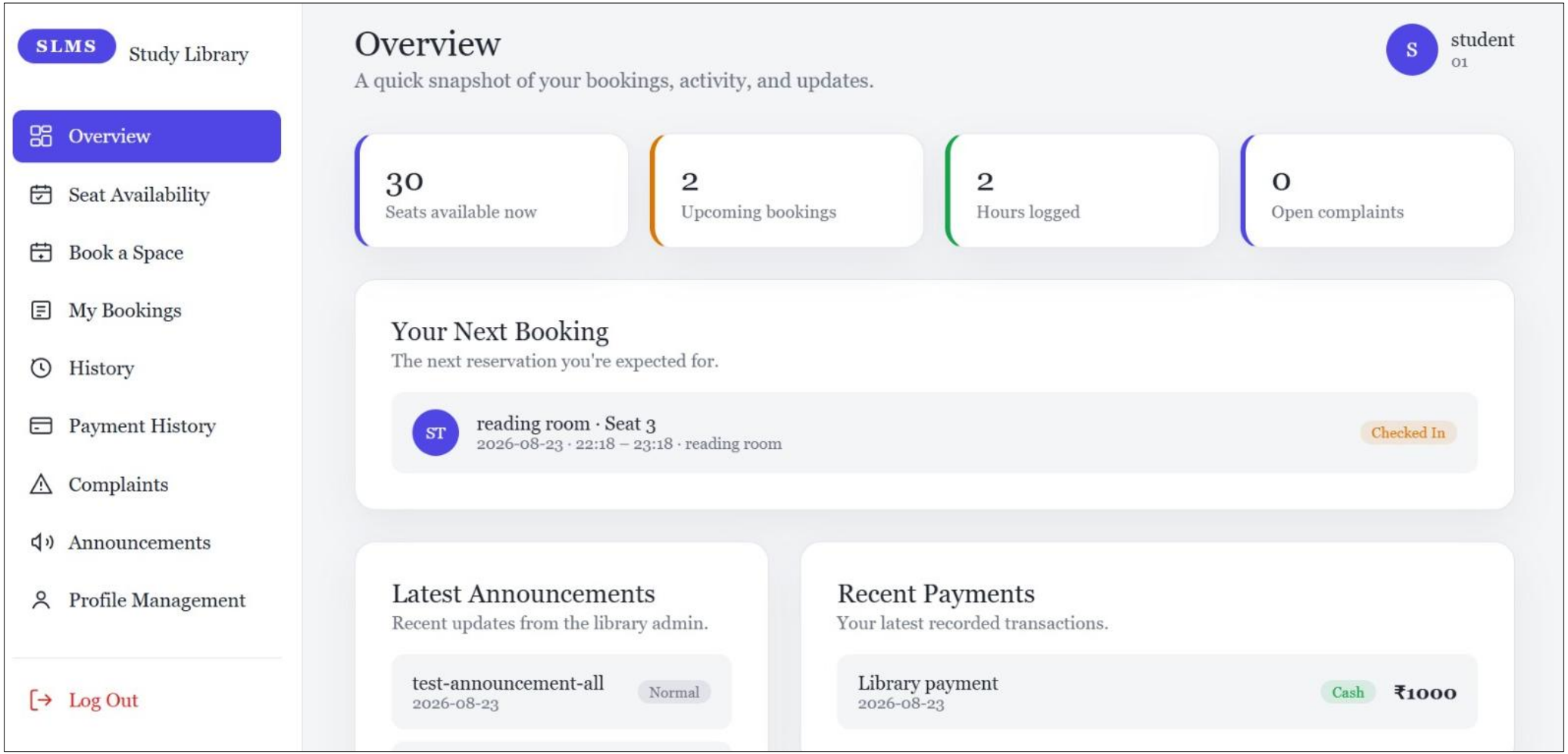


Fig: Student Dashboard

From Design to Working System

Requirements › System Design › Database Design › Frontend › Backend APIs › Authentication & Authorization › Integration › Testing › **Final System**

Role-Based Access Control

Valid credentials	Invalid credentials	Unauthenticated access	Wrong role	Invalid resource
HTTP 200 → Access token	HTTP 401	HTTP 401	HTTP 403	HTTP 404

Comprehensive API Testing

Coverage across every functional area of the platform.

Authentication
Login and invalid credentials

Administration
Staff · Plans · Settings · Seats

Students
Registration · Profile · Membership

Bookings
Availability · Create · Update · Cancel

Extensions
Request · Approve · Reject · Reallocate

Attendance
Check-in · Checkout · No-show

Payments
Record · History · Validation

Complaints
Submit · Review · Update

Communication
Announcements

Reporting
Occupancy · Membership statistics

Testing Beyond the Happy Path

ZenStudy validates real business conditions.

Wrong password 401	Missing authorization 401	Wrong role 403	Unknown student 404	Invalid extension time 400
Duplicate extension 409	Booking conflict 409	Invalid payment 400	Duplicate check-in 400	Duplicate checkout 400

Automated Testing with Pytest

Authentication · Authorization · Staff Management · Memberships · Seats · Bookings · Extensions · Attendance · Payments · Complaints · Announcements · Reports

Result: Core API workflows successfully validated through automated tests.

Technology Foundation

Frontend

Role-based responsive web interfaces

Backend

API-driven application architecture

Database

Relational database with structured entities and relationships

Authentication

Token-based authentication and role-based authorization

Testing

Pytest-based automated API testing

Project Management

Jira / Kanban · GitHub workflow

From Discovery to Final System

1. Project Initiation & Planning
2. User Research & Requirements
3. System Planning & Design
4. Frontend Development — Phase I
5. Frontend Development — Phase II
6. Backend Development & Integration
7. Testing & Quality Assurance
8. Final Documentation & Submission

Built Through Agile Collaboration

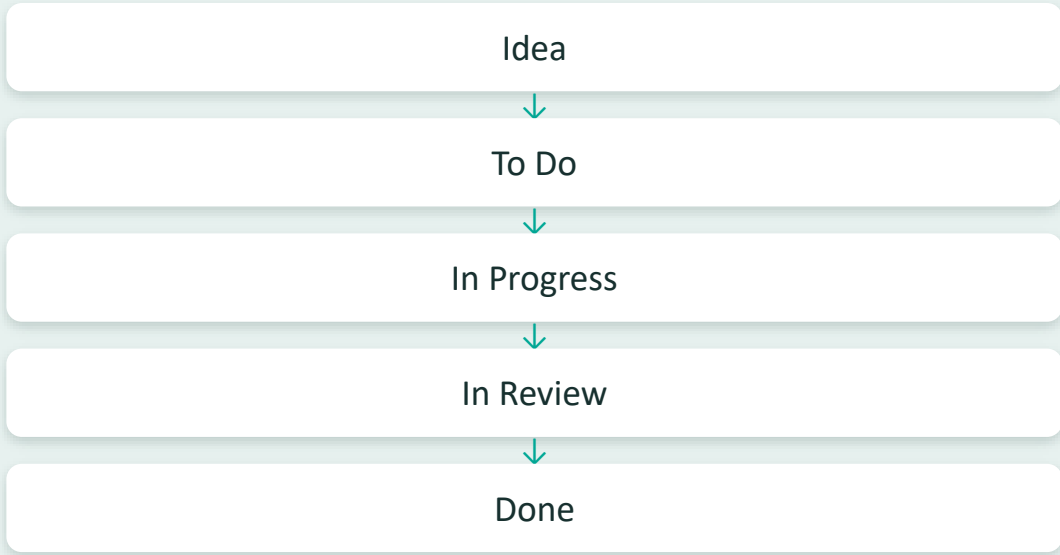
The project documentation records both the Scrum meeting process and Kanban workflow.

Scrum

Regular meetings for:

- Planning
- Progress review
- Requirement validation
- Design review
- Testing
- Refinement

Kanban



Result: A transparent workflow where every feature moved from idea to implementation and validation.

ZenStudy — The Completed Platform

- ✓ Student Management
- ✓ Membership Management
- ✓ Seat Availability
- ✓ Booking Management
- ✓ Booking Extensions
- ✓ Check-In / Check-Out
- ✓ No-Show Management
- ✓ Payment Management
- ✓ Complaint Management
- ✓ Announcements
- ✓ Administrative Configuration
- ✓ Occupancy Reporting
- ✓ Membership Analytics
- ✓ Role-Based Access
- ✓ Automated API Testing

Built. Tested. Refined.

Staff Management

Added improved filtering/search capability.

Membership Plans

Introduced the ability to manage plan availability without relying only on deletion.

Error Handling

Improved validation and user-facing error responses.

Integration

Completed the connection between frontend workflows and backend services.

What ZenStudy Delivers

Students

- Less administrative effort
- Better visibility
- Easier booking
- Structured study sessions

Library Staff

- Faster operations
- Better seat utilization
- Centralized records
- Easier issue resolution

Administrators

- Better control
- Better visibility
- Operational analytics
- Centralized configuration

One Platform. Better Operations.

ZenStudy transforms study-library management from fragmented manual workflows into a connected digital system.

Students

Focus on studying.

Staff

Focus on service.

Administrators

Focus on decisions.

Library

Focus on efficient utilization.

Beyond ZenStudy — Future Vision

- **Smarter Analytics**
- **Automated Notifications**
- **Advanced Mobile Experience**
- **Online Payments**
- **Predictive Capacity Planning**

Final Takeaways

01 A complete role-based study-library management platform.

02 Centralized membership, booking, attendance and payment workflows.

03 Digital seat allocation and no-show management.

04 Extension and complaint-management workflows.

05 Administrative configuration and reporting.

06 Secure role-based API access.

07 Comprehensive automated API testing.

08 A system refined through user feedback.

Thank You

ZenStudy · DeerInTheCode · Team-051